JEV: a lower-cost evaluation option for ADI

Pilot JEV for bounded scoring and classification; keep GPT-4o for complex review and explanations.

48.3x
Lower estimated token cost
JEV vs GPT-4o judge

3.2x
Higher dataset throughput
Same concurrency; batching differs

98.6%
JEV quality-label accuracy
GPT-4o: 94.6%; designer labels

WHAT WAS TESTED

35 synthetic ADI cases; 2 runs per judge.
Five quality metrics + a correctness check.
Concurrency 3; context cap 8,000 characters.

WHAT IS NOT ESTABLISHED

GPT-4o-mini and SLMs were not tested.
No proof of replacing answer generation.
No production uptime or compliance validation.

Decision requested: approve an independently labelled validation pilot, not a model replacement.

Source: comparison workbook / Summary and Setup. Synthetic evidence, not a production assessment.

Different model roles, not just different prices

JEV offers structured decisions; generative models retain broader language output and review capability.

Dimension	GPT-4o judge	GPT-4o-mini / SLM	JEV
Core approach	Generates JSON scores and written reasons	Generative judging; SLM depends on chosen model	Typed Score decisions; probabilities + confidence
Integration	Prompt + query, answer and retrieved context	Prompt / tuning; validation required	State + rubric questions; 5 quality metrics per call
Best-fit role	Complex review; explanations; escalation	Candidate low-cost judge or answer-generation model	Routine scoring; guardrail / routing candidates to test
Trade-off	Flexible; explanation cost; prompt / parsing sensitivity	Quality and cost depend on model, hosting and tuning	No generated rationale; rubric / threshold sensitivity
Evidence today	35 cases x 2 runs	Not benchmarked; no ranking supported	35 cases x 2 runs; model jev-1.13.0

SLM = small language model, not one specific model. Select the model and deployment before comparing it.

Sources: implemented evaluator APIs and workbook Setup. GPT-4o-mini / SLM column describes options, not results.

Same 0-1 range, different scoring mathematics

Quality metrics use the query, answer and context. Correctness additionally needs the SME's corrected answer.

Metric	GPT-4o: prompt-directed scoring	JEV: probabilistic rubric scoring
Faithfulness	Supported atomic claims / all factual claims. No factual claims -> 1.0.	4 support levels (0-3); expected level / 3.
Answer relevance	On-topic response statements / all response statements; ignore accuracy.	4 on-topic levels (0-3); expected level / 3.
Context relevance	Relevant retrieved-context statements / all retrieved-context statements.	4 retrieval-relevance levels (0-3); expected level / 3.
Toxicity	G-Eval raw 1-5 -> (raw - 1) / 4. Lower = safer.	4 severity levels (0-3); expected level / 3; lower = safer.
Tone	G-Eval raw 1-5 -> (raw - 1) / 4. Higher = more professional.	4 professional-tone levels (0-3); expected level / 3.
Correctness (GT required)	Semantic match to SME correction; 0 / 0.5 / 1 on applicable rows.	3 semantic-match levels (0-2); expected level / 2 on applicable rows.

JEV raw = sum(level x probability). Example: [0.10, 0.20, 0.30, 0.40] -> 2.00 / 3 = 0.667.

GPT ratios are model estimates, not independently counted claims. JEV confidence is separate from the score.

Sources: adi_gt_evaluator.py metric prompts; jev_judge.py rubrics and score normalization.

Five quality metrics without ground-truth answers

These metrics need no SME reference answer. Inter-judge agreement is not a measure of accuracy.

Quality metric	Better	GPT-4o mean	JEV mean	Pass/fail agreement
Faithfulness	Higher	0.7343	0.6853	77.1%
Answer relevance	Higher	0.9429	0.8929	100.0%
Context relevance	Higher	0.8714	0.8302	100.0%
Toxicity	Lower	0.0571	0.0376	97.1%
Tone appropriateness	Higher	0.9229	0.9195	100.0%

Faithfulness diverges most (77.1% agreement): review unsupported-claim disagreements.

Cut-off = 0.5; high toxicity fails. Excludes correctness, SME agreement, tag matches and document precision/recall.

Source: comparison workbook / Agreement. Means and verdict agreement averaged over 2 runs, 35 cases each.

Measured economics and reliability

35 rows, 2 runs, concurrency 3. Token estimates are not total cost; observed stability is not a service SLA.

Measure	GPT-4o judge	JEV
Cost per 1,000 evaluated rows	$2.228	$0.046
Wall time for 35-row dataset	14.98 s	4.68 s
Dataset throughput	2.34 rows/s	7.48 rows/s
Median API-call latency	4.113 s	0.346 s
p95 API-call latency	5.150 s	0.495 s
Failed calls / total observed (both runs)	0 / 20	0 / 70
Rows with missing scores / total (both runs)	0 / 70	0 / 70
Mean per-metric verdict flip rate	0.48%	0.48%
Mean absolute run-to-run score change	0.0121	0.0056

Batching differs: GPT-4o uses 5 rows/call + correctness calls; JEV uses 1 row/call. Prefer dataset throughput.

Rates per 1M tokens: GPT-4o $2.50 input / $10 output; JEV $0.042 input / $0 output. No uptime claim.

Sources: comparison workbook / Summary, Latency_Cost and Consistency. USD token estimates; same local test setup.

ADI recommendation: a confidence-gated pilot

Approve validation, not replacement.

JEV routine screening › **Low confidence / disagreement / high risk** › **GPT-4o or SME review**

Phase	Recommended action	Decision evidence
1. Shadow test	500+ de-identified, independently SME-labelled cases; include numbers, refusals and Hinglish.	Failure recall, correctness, latency; confidence slices and vendor data controls.
2. Bounded pilot	Start with routine quality scoring; keep high-risk answers in the escalation path.	Calibrate score / confidence cut-offs; retain audit logs and monitor drift.
3. Compare options	Test GPT-4o-mini + named SLMs on the same data, task definitions and operating budgets.	Per-metric recall, explanations, privacy, hosting, availability and total cost.

Retain GPT-4o for difficult correctness review: 80.0% vs JEV 73.3% band accuracy on 15 applicable synthetic cases. Human review remains necessary for high-risk banking decisions.

Recommendation, not a measured production architecture. Correctness figures: workbook Summary, 15 applicable cases.